

Week 3 Lab: VAE, Latent Diffusion and Flow Matching

Exercise 1

The Inception Score (IS) feeds each generated image y^n into an off-the-shelf classifier to get $P(c|y^n)$, and averages these to get $P(c) = \frac{1}{N} \sum_n P(c|y^n)$. Formally,

$$\text{IS} = \exp \left(\frac{1}{N} \sum_n \text{KL}(P(c|y^n) \| P(c)) \right).$$

- According to the slides, what should $P(c|y)$ look like for a high-*quality* image, and what should $P(c)$ look like for a *diverse* set?
- Compute IS for three images with classifier outputs (0.9, 0.05, 0.05), (0.05, 0.9, 0.05) and (0.05, 0.05, 0.9). Then recompute when the second image is replaced by a copy of the first. Interpret.
- Name one failure mode that IS cannot detect.

Exercise 2

FID embeds real and generated images with the Inception network, fits a Gaussian to each set of embeddings and reports the Fréchet distance between the two Gaussians. In 1-D this distance is $(\mu_r - \mu_g)^2 + (\sigma_r - \sigma_g)^2$.

- Real features are $\mathcal{N}(0, 1)$. Compute the 1-D FID for generated features $\mathcal{N}(0.5, 1)$, $\mathcal{N}(0, 0.25)$ (std 0.5), and $\mathcal{N}(0, 1)$.
- Which of these corresponds to mode collapse? Is a smaller or a larger FID better?
- Why does FID need “a lot of samples”, and why is it still not enough to detect a model that simply copies training images?

Exercise 3

For text-to-image models we also want to know whether the image matches the prompt (“A running dog”).

- How is CLIP trained on its 400 million image-text pairs, and how is it used to score a generated image?
- Give one situation in which a high CLIP score does not mean a good image.

Exercise 4

In the lecture’s VAE the encoder outputs m_i and σ_i , samples $e_i \sim \mathcal{N}(0, 1)$, and forms the code $c_i = \exp(\sigma_i) e_i + m_i$. Besides the reconstruction error it minimises $\sum_i (\exp(\sigma_i) - (1 + \sigma_i) + m_i^2)$.

- Why does decoding a *randomly generated* code usually fail for a plain auto-encoder?
- What happens if the VAE only minimises the reconstruction error? (Consider what the encoder would do with σ_i .)
- Show that the extra term is minimised at $\sigma_i = 0$, $m_i = 0$. What does this mean for the distribution of codes? Which part acts like L2 regularisation?

Exercise 5

In the lab both models have a 2-D latent space; you plot the codes of held-out digits and decode a grid inside a dashed square (the region where most mass of $\mathcal{N}(0, I)$ lies).

- Where would you expect AE codes and VAE posterior means to lie relative to the square, and where does each decoded grid produce recognisable digits?

- (b) Must the VAE's KL curve decrease every epoch? Explain the trade-off between the two terms.

Exercise 6

Flow-matching training pairs a noise point $x_0 \sim \mathcal{N}(0, I)$ with a data point x_1 , samples $t \sim U(0, 1)$, forms $x_t = (1 - t)x_0 + tx_1$, and regresses the network output $\hat{v} = v_\theta(x_t, t)$ onto $v = x_1 - x_0$ with MSE.

- (a) For $x_0 = (1, -1)$, $x_1 = (3, 2)$ and $t = 0.25$, compute the network input x_t and the target v . Show that $v = \frac{d}{dt}x_t$.
- (b) Two different pairs can pass through (almost) the same (x_t, t) with different targets v . What does the MSE-optimal network predict there? Are the learned sampling trajectories straight?
- (c) Compare with the diffusion training loop (L3 p. 39): what is the network asked to predict in each case?